

Death by Survey

Estimating Adult Mortality without Selection Bias

Emmanuela Gakidou Gary King

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Sources of Mortality Data

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(death certificates, mostly in developed countries)
- Demographic surveillance systems
(a few isolated projects)
- Wild guesses
(typically reported in the media, usually citing other wild guesses as authorities)

Who Uses Mortality Data?

- Medical researchers

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- Public health researchers

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 - Should also study: more ultimate outcomes, like human misery or mortality

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- Political scientists:
 - IR studies: the elite decision to go to war
 - Should also study: more ultimate outcomes, like human misery or mortality
 - The **Big Problems**: uncertainty and selection bias

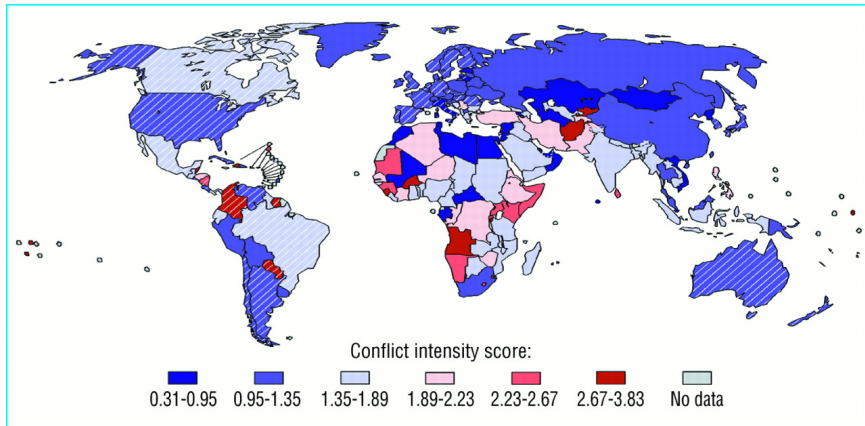
Uncertainty in Death Estimates from Major Sources

The 1990s' Ten Most Deadly Conflicts

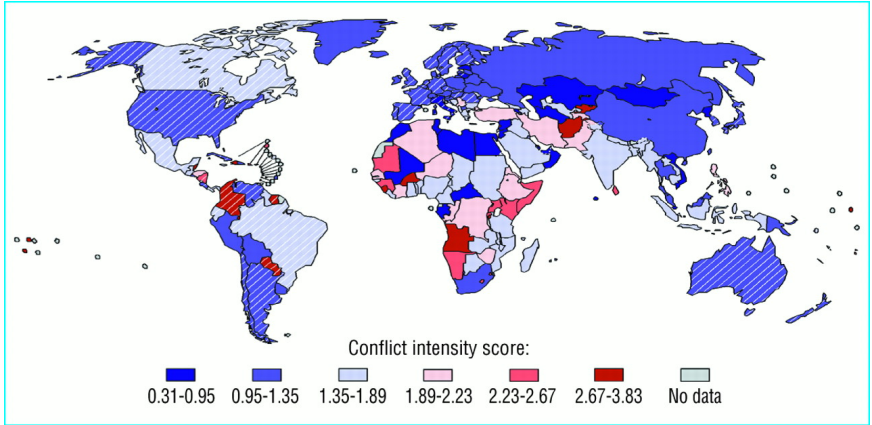
Country	Year	People Killed (in 1000s)		
		Low	High	Range
Rwanda	1994	500	1,000	500
Angola	1992–4	100	500	400
Somalia	1991–9	48	300	252
Bosnia	1992–5	35	250	215
Liberia	1991–6	25	200	175
Burundi	1993	30	200	170
Chechnya	1994–6	30	90	60
Tajikistan	1992–9	20	120	100
Algeria	1992–9	30	100	70
Gulf war	1990–1	4.3	100	95.7

Selection bias in Mortality from War Data

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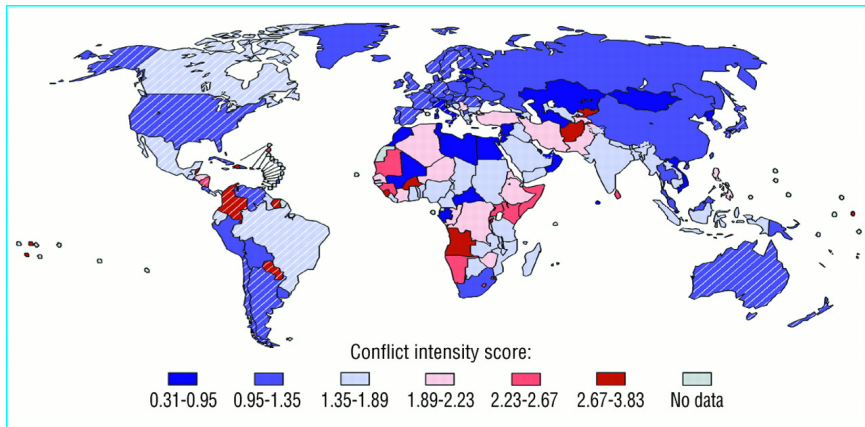


Selection bias in Mortality from War Data



Vital registration areas (cross-hatched): mostly low conflict

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Vital registration areas (cross-hatched): **mostly low conflict**

No registration areas: **much higher conflict**

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- **Spouses** (Malaker 1986; Stanton, Nouredine and Hill 2000; Singh 2000; Timaeus 1991)
- **Household residents and others** (Feeney 2001; Graham, Brass, and Snow, 1989)

Extensive Efforts to Develop and Apply These Methods

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Several hundred articles published in:

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- Public health
- Medicine
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- Demographic and Health Surveys (70 countries)

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Methodology

The Tradeoff

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- Joining Demography and Statistics (through political science)
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- Joining Demography and Statistics (through political science)
- Directing public health dollars (9% of the world's economy)
- Understanding the consequences of war and the causes of war

A Simplified Quantity of Interest

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Mortality =

$q =$

d_j	Sibship
0	1
1	1
0	1
1	2
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- Applied as estimators to (time 1) samples: unbiased

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Vital registration data

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Data available to sample at time 2

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- Each respondent reports on entire sibship
- High mortality sibships: underrepresented
- No sibships with 0 survivors
- Some families counted more than once

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- Sample includes n “survivors” by design

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A little miracle occurs

Trussell and Rodriguez (1990) prove: **if mortality is independent of sibship size**, all biases cancel: $\check{q}$ is unbiased.

Is mortality independent of sibship size?

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Peru	2000	Guinea	1999
Indonesia	1997	Zimbabwe	1994
Burkina Faso	1998	Nepal	1996
Benin	1996	Cameroon	1998
Peru	1996	Cote D'Ivoire	1994
Nigeria	1999	Togo	1998
Philippines	1998	Eritrea	1995
Chad	1997	Ethiopia	2000
Brazil	1996	Zimbabwe	1999
Indonesia	1994	Colombia	1995
Senegal	1999	Zambia	1996
Philippines	1993	Uganda	1995
Mali	1996	Madagascar	1997
Tanzania	1996		

Is mortality independent of sibship size? No!

Correlation			Correlation		
Peru	2000	0.97	Guinea	1999	0.80
Indonesia	1997	0.96	Zimbabwe	1994	0.76
Burkina Faso	1998	0.95	Nepal	1996	0.75
Benin	1996	0.95	Cameroon	1998	0.75
Peru	1996	0.95	Cote D'Ivoire	1994	0.75
Nigeria	1999	0.93	Togo	1998	0.74
Philippines	1998	0.93	Eritrea	1995	0.70
Chad	1997	0.93	Ethiopia	2000	0.71
Brazil	1996	0.92	Zimbabwe	1999	0.69
Indonesia	1994	0.91	Colombia	1995	0.52
Senegal	1999	0.90	Zambia	1996	0.47
Philippines	1993	0.88	Uganda	1995	-0.06
Mali	1996	0.86	Madagascar	1997	-0.19
Tanzania	1996	0.82			

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- Two problems addressed separately:
 - ① Underrepresentation of high mortality families
 - ② Nonrepresentation of families with zero survivors

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(Temporarily assuming no families with 0 survivors)

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- Sampling sibships at time 1: proportional to "Births" (B_i)

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- Sampling sibships at time 2: proportional to Survivors (S_i)

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- Sampling sibships at time 1: proportional to "Births" (B_i)
- Sampling sibships at time 2: proportional to Survivors (S_i)
- Estimation at time 1: simple average is unbiased

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- Estimation at time 1: simple average is unbiased
- Estimation at time 2:

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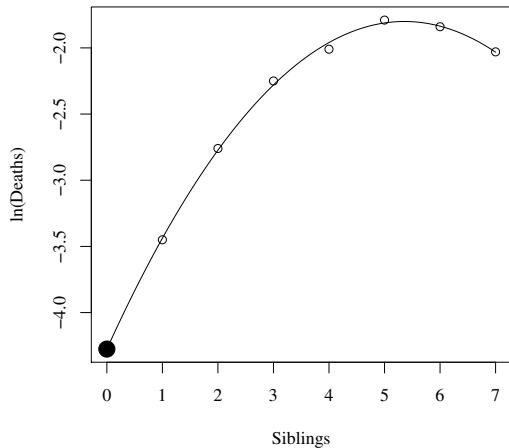
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- Thus, this part of the answer is more uncertain

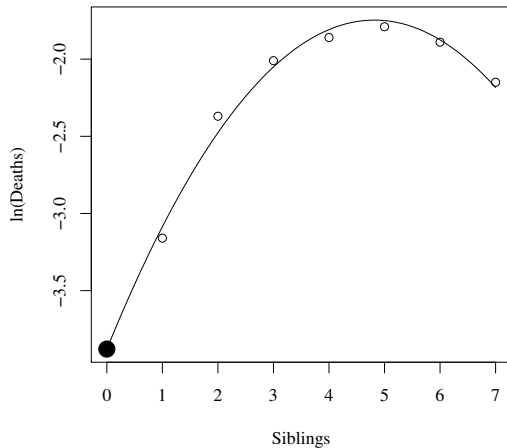
Extrapolation to Deaths in Families with 0 Survivors

Cote D'Ivoire 94



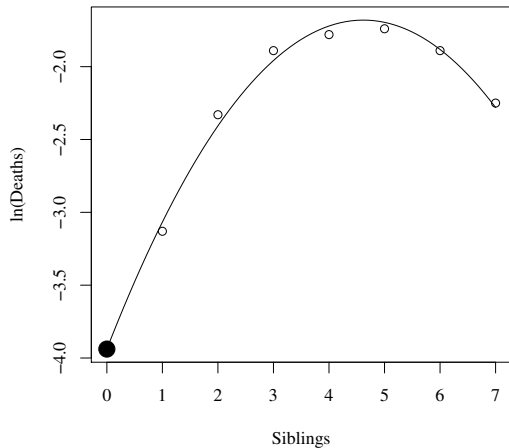
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Indonesia 94

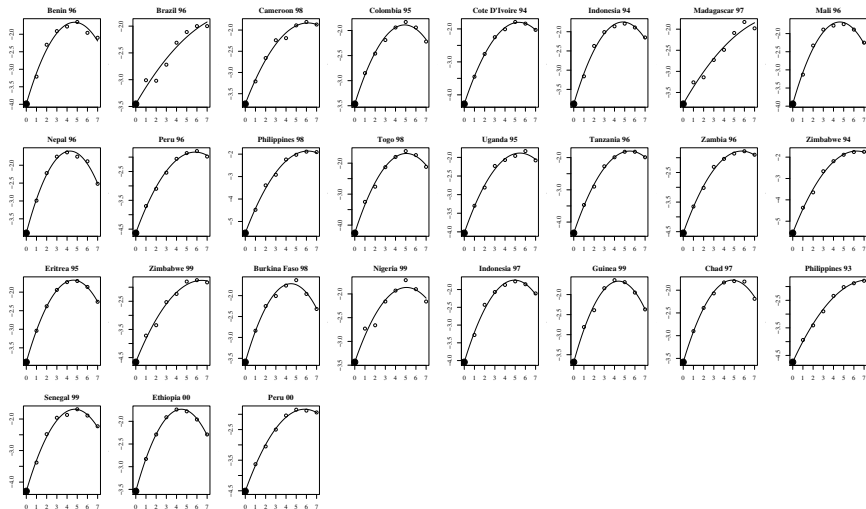


Extrapolation to Deaths in Families with 0 Survivors

Mali 96



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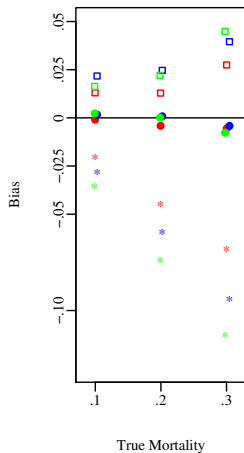
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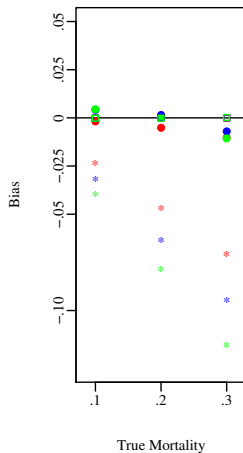
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For each, create 1,000 data sets, each with $n = 1,000$ randomly drawn time 2 survey respondents

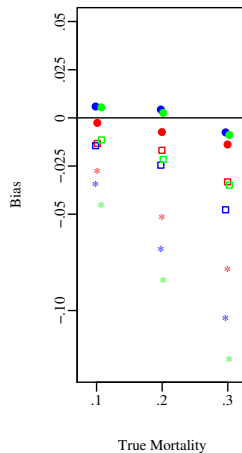
Corr(B,M) > 0



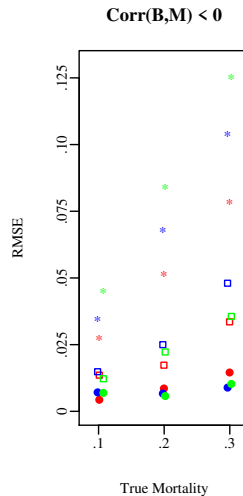
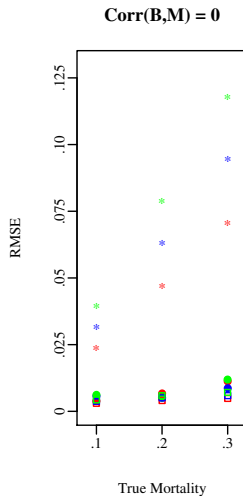
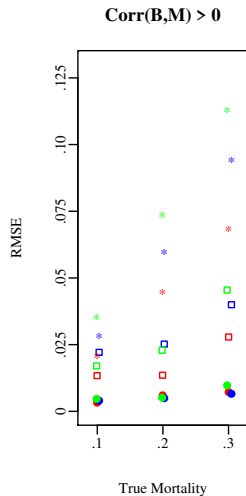
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Root Mean Square Error



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- Weighting plus extrapolation for families with 0 survivors: 92% corrected

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- Thus, an equivalent expression for $\hat{q}$:

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- ⑤ Ideally, a new subfield within IR predicting mortality and human misery with war and the predictors of war.

<http://GKing.Harvard.edu>